

# Kevin Chang

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## EDUCATION

### University of Michigan

*Master's of Science in Engineering*

**Program:** Electrical & Computer Engineering M.S.E. (Computer Vision)

Ann Arbor, MI

*Expected Graduation: December 2026*

*GPA: 4.00/4.00*

### University of Michigan

*Bachelor's of Science in Engineering*

**Major:** Computer Science

**Minor:** User Experience Design

Ann Arbor, MI

*Graduated: December 2025*

*GPA: 3.78/4.00 (Summa Cum Laude)*

## EXPERIENCE

### Rocket Close (Rocket Family of Companies)

*Software Engineer Intern*

Detroit, Michigan

*May 2026 - August 2026*

- Built the majority of admin-facing screens for a **full-stack** rewrite of Inquire, an internal Power BI/Data Explorer reporting platform used by thousands of business members and team leaders across the Rocket Family of Companies, using **.NET Core**, **PostgreSQL**, and **Angular**.
- Implemented end-to-end metadata management features, including ownership assignment, security group permissions, banner notices, and report discovery/search, while working across multiple Angular versions during a framework upgrade.
- Owned the redesign of report tagging end-to-end, migrating tags from plain strings on legacy report objects to first-class relational entities. Designed the **PostgreSQL** schema and migration scripts, built the **.NET Core backend API**, and refactored the **AWS Lambda** sync pipeline to diff and reconcile tags between legacy SQL Server and the new PostgreSQL model, resolving a duplicate-insert bug via pre-fetch/name-matching, then built the Angular UI on top of it.
- Ran **CI/CD** deployment pipelines in **Azure DevOps** and contributed **Terraform** changes to **AWS** infrastructure.
- Presented a live demo of Inquire to Rocket Close's Data Engineering department, covering platform architecture and metadata management capabilities.

### United Wholesale Mortgage

*Software Development Intern*

Pontiac, Michigan

*May 2025 - August 2025*

- Initiated migration of legacy **CRON jobs** to **C# Azure Functions**, completing 4 during the internship and laying the foundation for a cloud-native scheduling system.
- Built and tested backend solutions using **Kafka**, **Redis**, and **Dynatrace**, with unit tests written in **xUnit** to ensure reliability and maintainability.
- Rotated through Developer, QA, BA, and PO roles across 6 **Agile** sprints.
- Presented a live demo of an internal enterprise platform to 1,000+ employees.

### University of Michigan, Liu Lab

*Software Developer / Researcher*

Ann Arbor, Michigan

*March 2025 - Present*

- Designed and implemented a multi-agent **reinforcement learning** system that trains LLM Cypher-generation agents through execution-feedback rewards, enabling a 14B-parameter model to reach parity with 200B+ parameter models on **knowledge graph** query accuracy benchmarks.
- Led deployment and development of research web tools in collaboration with Dr. Shuibing Chen's lab at Weill Cornell Medicine, transitioning legacy research platforms to modern containerized systems using **Docker**, **React**, and **FastAPI** frameworks and integrating **R-based visualizations** for data analysis.
- Built benchmarking pipelines, custom **MCP** server tooling, and a **System 2 deep research** pipeline for multi-model LLM evaluation, supporting the GLKB paper accepted at KDD '26.
- Integrated the GLKB LLM agent into an automated biomedical **literature synthesis** pipeline, achieving a 36% improvement in extraction accuracy and precision over prior baseline systems.

#### Publications:

- Huang, Y., Han, Z., Chang, K., Jiao, T., Liu, J. Enhancing Biomedical AI Foundations: Genomic Literature Knowledge Base Boosts LLMs' Mastery of Biomedical Literature. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '26)*, Vol. 2, 9149–9158. ACM, 2026. <https://doi.org/10.1145/3770855.3817461>

#### Publications (under review):

- *Unraveling Human Sinoatrial Node Development Using Fetal Heart and SAN-Paced Cardioid Models*
- *Integrative Single-Cell Spatial Multi-Omics Analysis of Human Fetal Intestine*

## SKILLS

**Languages:** C++, C#, Python, JavaScript, TypeScript, SQL, Cypher, C, Bash, Verilog

**Frameworks/Libraries:** React, Angular, .NET Core, Dapper, FastAPI, Flask, NumPy, PyTorch, Pandas, Matplotlib, xUnit

**Databases:** PostgreSQL, SQL Server

**Tools & Platforms:** Linux, Docker, AWS (Lambda), Azure, Terraform, Git, Dynatrace, Redis, Kafka